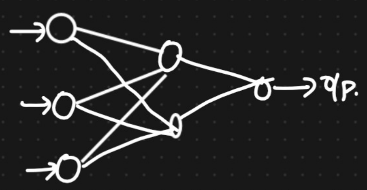
NLP In Deep Learning

① ANN → ARTIFICIAL NEURAL N/W → Tabular Data.

House size   No. of Rooms   Price



order is Important

Sequential Data
vs
Non Sequential Data

② CNN → Images → Image classification, Object Detection

Sequence of Data Is Important

③ i) RNN ⇒ Sequential Data. [NLP] [Time Series]
ii) LSTM RNN
iii) GRU RNN
iv) Encoder Decoder
v) Attention is all you need

Eg: Chat bot App → Q & A

Language Translation → Eng → French

Text Generation → A sentence → Completion of sentences

Auto Suggestion → LinkedIn, Gmail

Time Series → Sales Data Future Prediction.

① Can we solve With ANN?

Eg:

Text	O/P
The food is good	1
The food is bad	0
The food is not good	0.

① out of vocabulary

Words → Vectors

⇓ Sequential Info.

Bow, TF-IDF [Context Is Missing].

	Word 1	Word 2	Word 3	-	-
Sent 1	0.	1	0	0	0
Sent 2	1	1	0	0	0
Sent 3	0	0	1	0	0

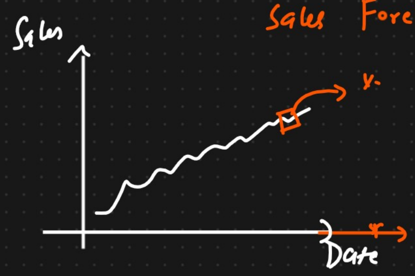
{good The food is} → Meaning may change

③ Language Translation → English   French
→ - - - -   → - - - - - -

④ Auto Suggestion → LinkedIn, GMAIL → Autosuggestion

Sales Forecasting

⑤ Sales Data → DateTime



Can we use ANN to solve this Problem? → Sequential Data

↓

NLP In Deep Learning   [Generative AI → LLM, MultiModel]

① Simple RNN → LSTM / GRU RNN → Bidirectional RNN → Encoder Decoder
↓
Transformers ← Self Attention

Can we Solve With ANN → Sequential Data

Dataset {Sentiment Analysis}   ANN

Text Preprocessing → Text → Vectors

Text	O/p
The food is good	1
The food is bad	0
The food is not good	0

Vocabulary Size → 4

RNN

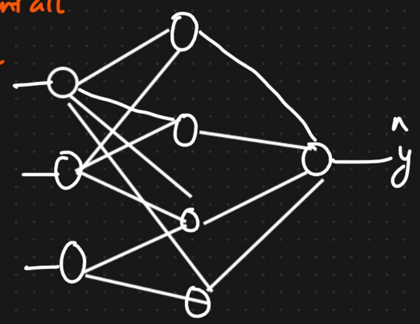
1. Timestamps
2. In/p Text·

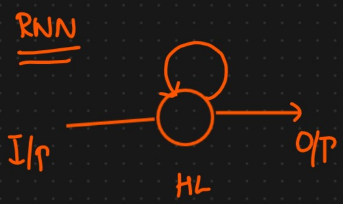
Text	O/p
The food is good	1
The food is bad	0
The food is not good	0·

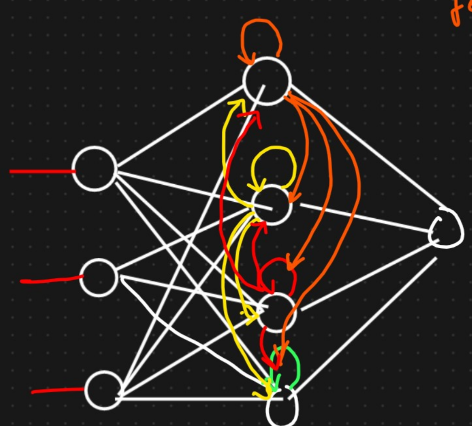
RNN Architecture

[ANN]

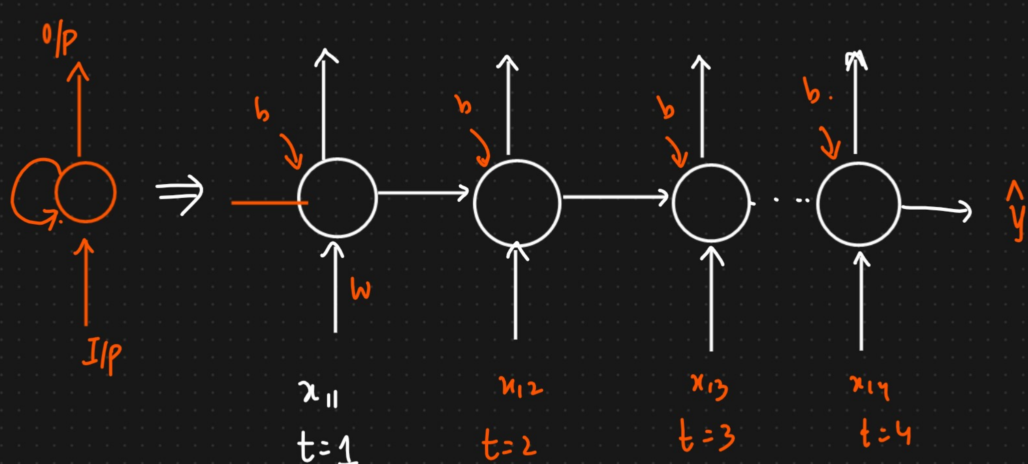
I/p is sent all at once





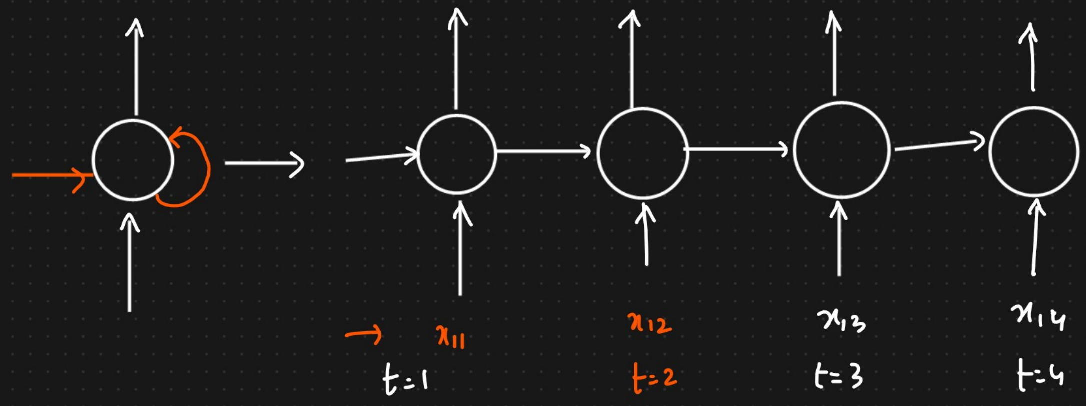


→ The food is good



Sentence

Words



Working Of Simple RNN With Forward Propogation

Dataset

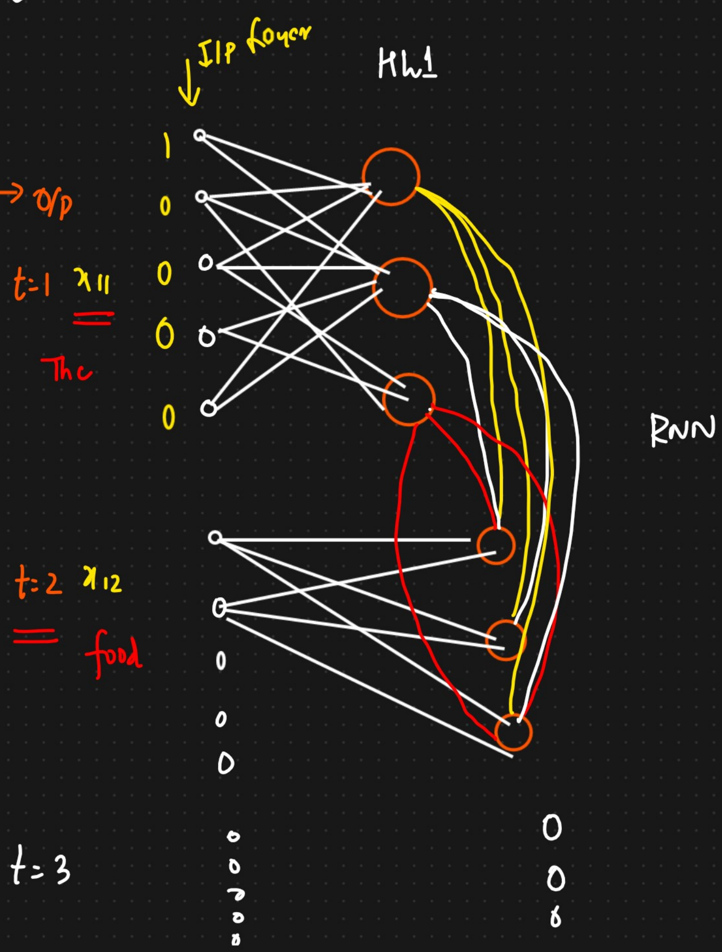
Text	O/P
The food is good	1
The food is bad	0
The food is not good	0

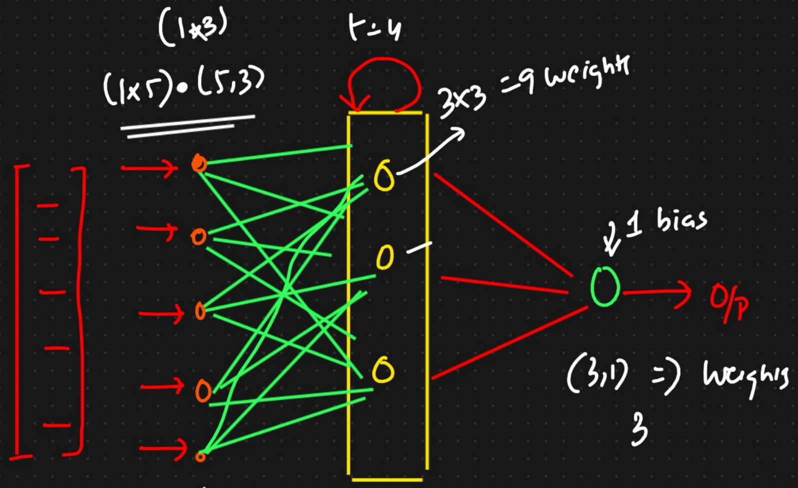
x_{11} = The, x_{12} = food, x_{13} = good

Words → Vectors

OHE → One hot Encoding

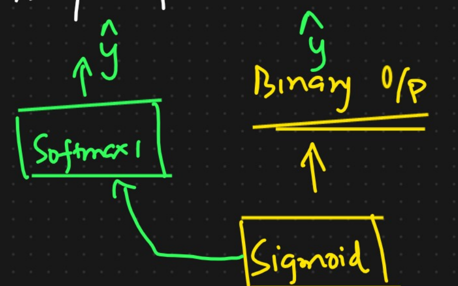
[[1 0 0 0 0] [0 1 0 0 0] [0 0 1 0 0] [0 0 0 1 0] [0 0 0 0 1]]

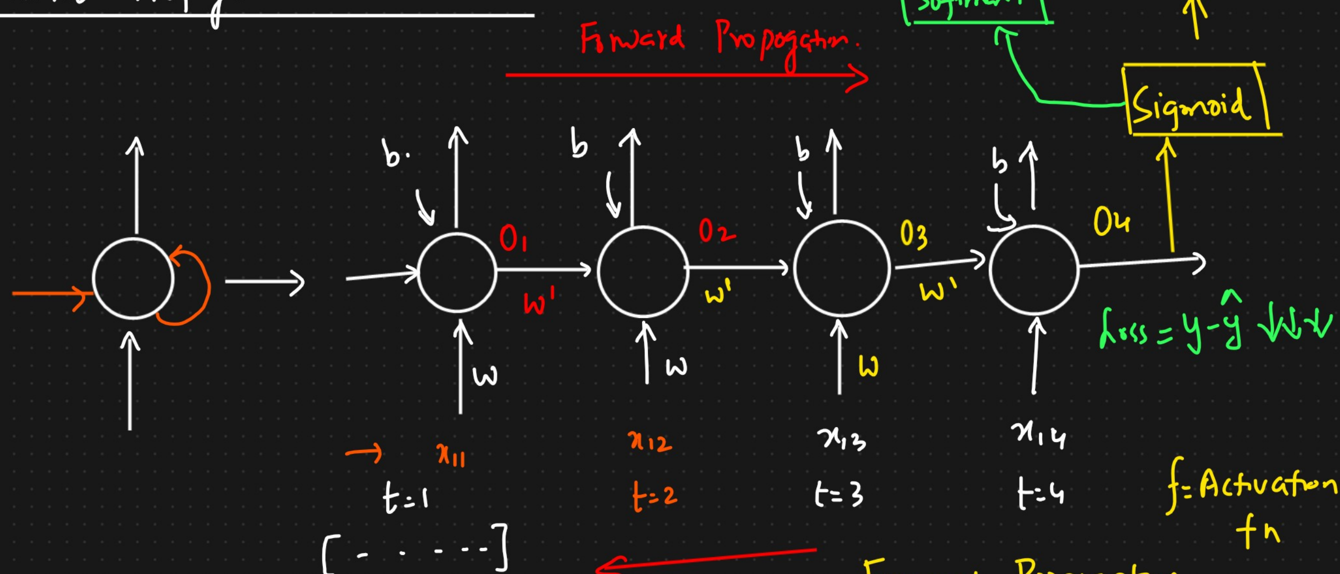




$27 + 4 = 31$ {Trainable param}.

Forward Propogation With Time





f: Activation fn

Forward Propogation

$$O_1 = f(x_{11} \cdot w + b_1)$$

$$O_2 = f(x_{12} * w + O_1 * w' + b_1)$$

$$O_3 = f(x_{13} * w + O_2 * w' + b_1).$$

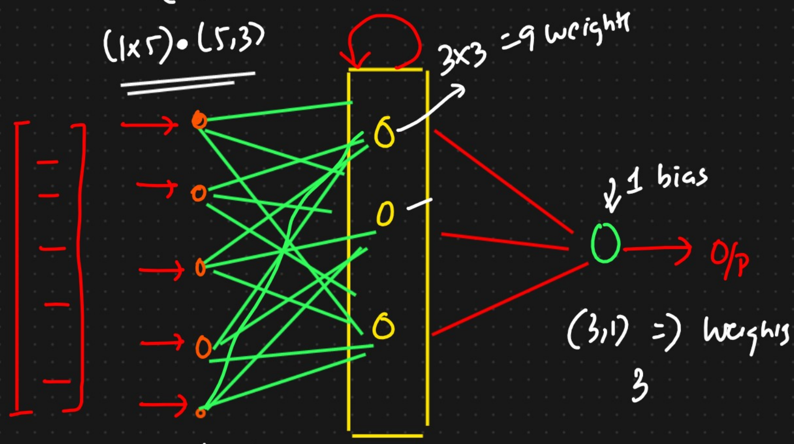
Dataset

	O/P
The food is good	1

x_{11} x_{12} x_{13} x_{14}.

(1x3)

(1x5)·(5,3)



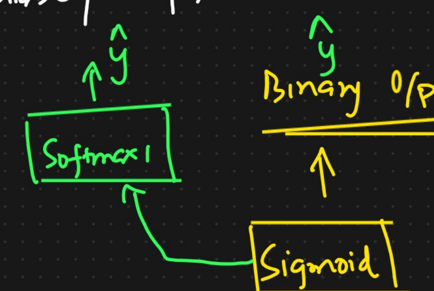
(5x3)

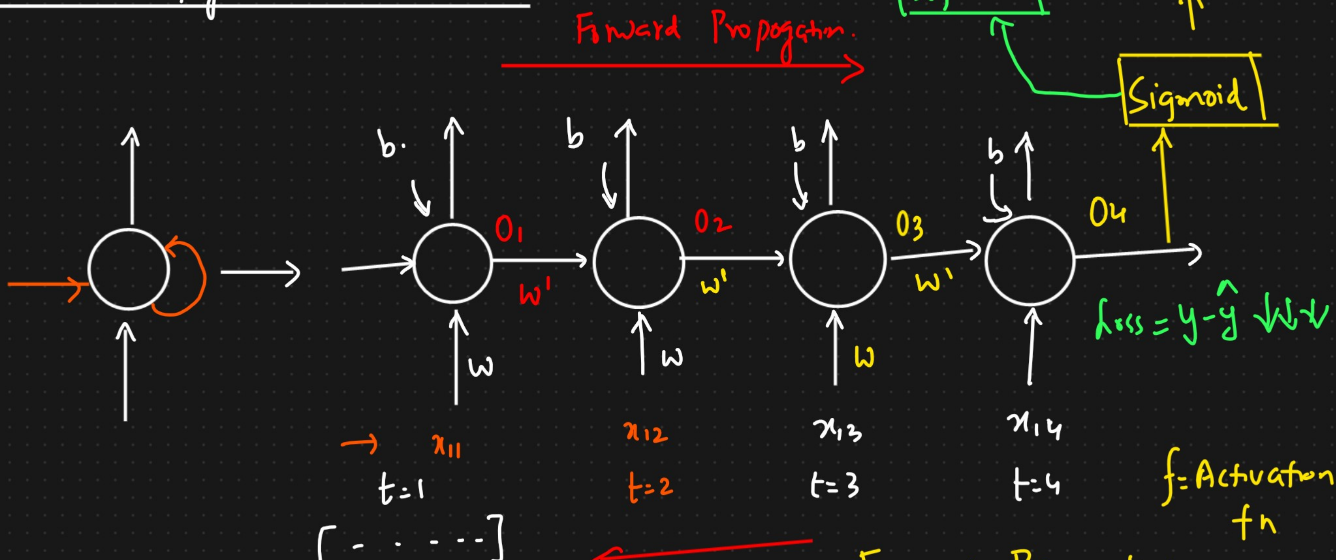
=15 weights

3 bias

27+4= 31 {Trainable param}.

Forward Propogation With Time





f = Activation fn

Forward Propogation

$O_1 = f(x_{11} \cdot w + b_1)$

$O_2 = f(x_{12} * w + O_1 * w' + b_1)$

$O_3 = f(x_{13} * w + O_2 * w' + b_1).$

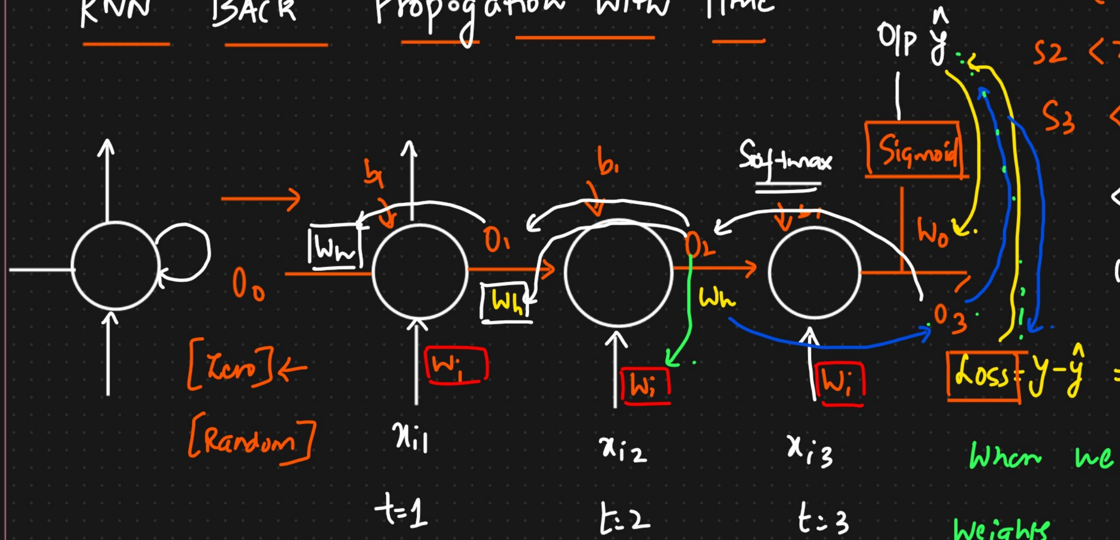
⋮

O_n

Dataset

	O/P
The food is good	1
x_{11} x_{12} x_{13} x_{14}.	

RNN BACK Propogation With Time



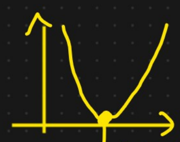
				O/p
S1	< x_{11}	x_{12}	x_{13} >	1
S2	< x_{21}	x_{22}	x_{23} >	0
S3	< x_{31}	x_{32}	x_{33} >.	

< x_{i1} x_{i2} x_{i3} >.

i = Sentence No.

When we update all the weights

[W_i, W_h, W_o] ← Back Propogation



↳ global Minima → loss ↓↓↓

Forward Propogation

$$O_1 = f(x_{i1} * W_i + O_0 W_h + b_1)$$

$$O_2 = f(x_{i2} * W_i + O_1 * W_h + b)$$

$$O_3 = f(x_{i3} * W_i + O_2 * W_h + b).$$

$$\hat{y} = \sigma(O_3 * W_o).$$

Backward Propagation with Time

Update [W_i, W_h, W_o]

Weight updation formula

$$W_{new} = W_{old} - \eta \boxed{\frac{\partial L}{\partial W_{old}}}$$

→ Derivative, slope of GRADIENT DESCENT

η = 0.001

$$\boxed{W_{o_{new}} = W_{o_{old}} - \eta \frac{\partial L}{\partial W_{o_{old}}}}$$

① Update W_o

Chain Rule of Derivative

$$\frac{\partial L}{\partial W_{o_{old}}} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial W_{old}}$$

② Update W_h. [Hidden Layer Weights] → Time Stamps

$$W_{h_{new}} = W_{h_{old}} - \eta \boxed{\frac{\partial L}{\partial W_{h_{old}}}}$$

t=1, 2, 3

$$\Rightarrow \frac{\partial L}{\partial w_{h\,old}} = \overbrace{\left[\frac{\partial L}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial o_3} * \frac{\partial o_3}{\partial w_h}\right]}^{t=3} + \overbrace{\left[\frac{\partial L}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial o_3} * \frac{\partial o_3}{\partial o_2} * \frac{\partial o_2}{\partial w_h}\right]}^{t=2}$$

$$+ \overbrace{\left[\frac{\partial L}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial o_3} * \frac{\partial o_3}{\partial o_2} * \frac{\partial o_2}{\partial o_1} * \frac{\partial o_1}{\partial w_h}\right]}^{t=1}$$

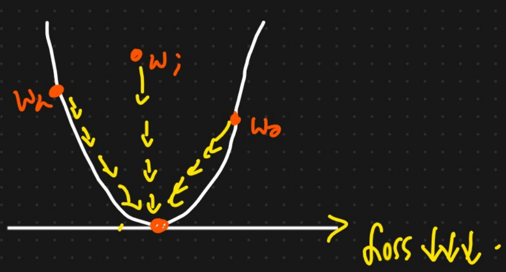
③ Updating Weights $W_i \longrightarrow$ Timestamp

$$W_{i\,new} = W_{i\,old} - \eta \boxed{\frac{\partial L}{\partial w_{i\,old}}}$$

{update W_i weights}

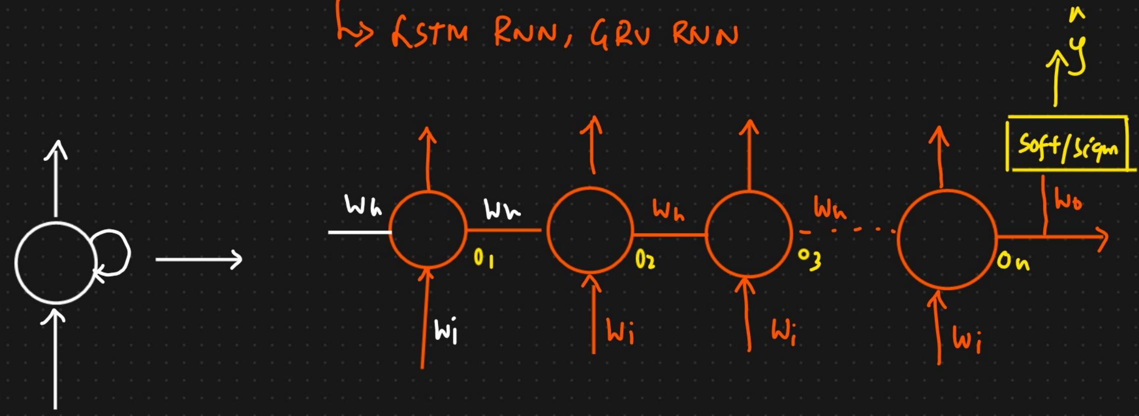
$$\boxed{\frac{\partial L}{\partial w_{i\,old}}} = \overbrace{\left[\frac{\partial L}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial o_3} * \frac{\partial o_3}{\partial w_{i\,old}}\right]}^{t=3} + \overbrace{\left[\frac{\partial L}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial o_3} * \frac{\partial o_3}{\partial o_2} * \frac{\partial o_2}{\partial w_{i\,old}}\right]}^{t=2}$$

$$+ \overbrace{\left[\frac{\partial L}{\partial \hat{y}} * \frac{\partial \hat{y}}{\partial o_3} * \frac{\partial o_3}{\partial o_2} * \frac{\partial o_2}{\partial o_1} * \frac{\partial o_1}{\partial w_{i\,old}}\right]}^{t=1}$$



Problems With RNN

↳ LSTM RNN, GRU RNN



$Loss = (y - \hat{y})$ ↓↓↓

ANN → Vanishing Gradient Problem.

Text	O/p
The food is good	1
The food is bad	0

Sentences → small

Text Generation

S1 → I like to play ___

food .

Videos

{ Sentence is . Long }

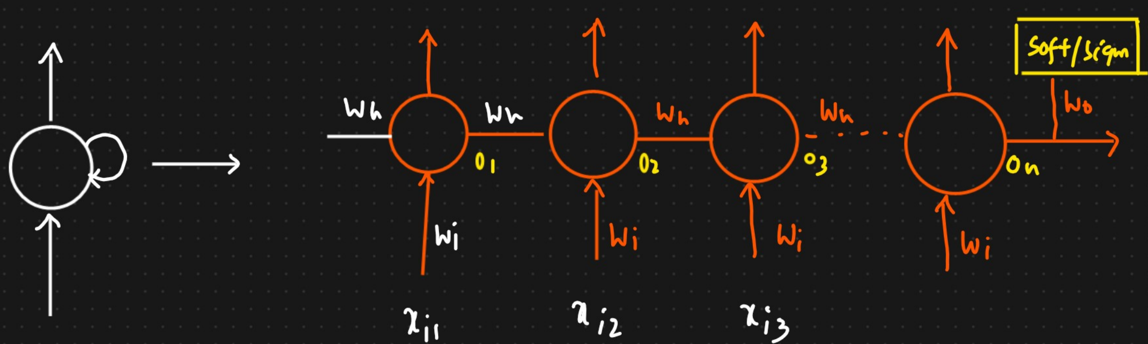
t=1 t=2 t=3 t=4 ?

My Name is KRISH and I like sports like CRICKET, VOLLEYBALL AND ALSO LIKE TO MAKE ___

t=16

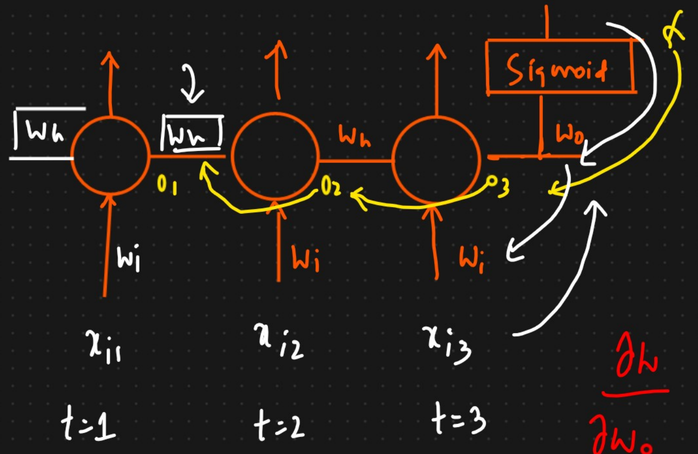
100 words

① Long term Dependency cannot be captured by RNN ↗ Provide Accuracy ⇓ Dependency.



$Loss = (y - \hat{y})$ ↓↓↓

$\hat{y}$ $Loss = (y - \hat{y})$



Updates W_h, W_i, W_o

$$\boxed{W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}}$$

$$\frac{\partial L}{\partial W_o} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial W_o} \Leftarrow$$

$t=1$ $\qquad\qquad$ $t=2$

$$\frac{\partial L}{\partial W_{h_{old}}} = \underbrace{\left[\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_3} \cdot \frac{\partial O_3}{\partial O_2} \cdot \frac{\partial O_2}{\partial W_{h_{old}}}\right]}_{} +$$

Length of sentence $\Rightarrow$ 50 words

$t=1$ $\qquad\qquad$ $t=2$

$0.25 \times 0.02 \times 0.01 \times 0.015$

$$\frac{\partial L}{\partial W_{h_{old}}} = \left[\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_{50}} \cdot \frac{\partial O_{50}}{\partial O_{49}} \cdot \boxed{\frac{\partial O_{49}}{\partial O_{48}}} \cdots \boxed{\vdots} * \frac{\partial O_2}{\partial W_{h_{old}}}\right] +$$

$\Downarrow$ $\qquad\qquad$ $\Downarrow$

Small Value ≈ 0

$$\frac{\partial O_3}{\partial O_2} = \frac{\partial \sigma(x_{i3} * W_i + O_2 * W_h + b)}{\partial O_2} \Leftarrow O_3.$$

$O_3 = \sigma(x_{i3} * W_i + O_2 * W_h + b)$

Derivative of sigmoid $0 - 0.25$

Vanishing Gradient Problem

$$= \sigma'(1 * W_h) \Rightarrow [0 - 0.25] * (W_h)$$

$t=1 \Rightarrow$ The word is not participating to update the weights value.

Chain Rule is Big

$t=50$ $\qquad$ $t=49$ $\qquad$ $t=48$

$$\frac{\partial L}{\partial w_{hold}} = \boxed{\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_{50}} * \frac{\partial O_{50}}{\partial w_{hold}}} + \boxed{} + \boxed{}$$

$\boxed{\approx 0}$

$\boxed{t=2}$ $\qquad$ $\boxed{t=1}$

$$\approx 0. \quad + \boxed{} \approx 0$$

① Relu, Leaky Relu →

② LSTM RNN ⟶ Long Short term Memory RNN } ⇒ Simple RNN.

③ GRU RNN ⟶